

Insert Your Project Title Here

Differential Equations Course Project

Student Name

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Abstract

Write approximately 100–150 words that summarize the completed project. A strong abstract briefly states (1) the application, (2) the research question, (3) the differential-equation model, (4) the main type of analysis or simulation, and (5) the main conclusion.

Do not write the abstract as an introduction. It should summarize the entire completed project.

1 Introduction and Research Question

Introduce the real-world problem or application. Explain why the topic is interesting or important and clearly state the main research question that your project investigates.

A useful research question is focused enough that your model and simulations can provide a meaningful answer. Examples include:

How does changing the parameter α affect the long-term behavior of the model?

or

How does adding an intervention term change the peak value of the population being studied?

At the end of this section, explicitly state your research question.

Research Question: *Insert your question here.*

2 Background

Provide enough scientific, social, engineering, or mathematical background for a reader to understand the application. Summarize relevant information from credible sources.

If you are adapting an existing mathematical model, explain where the model comes from and cite the source. Do not copy the language of a research paper. Explain the main ideas in your own words and cite the original source.

For this course project, approximately three or more credible references are normally appropriate, including at least one scholarly source.

3 Model Formulation

This is one of the most important sections of the report. Do not begin by simply displaying equations. First explain how the model is constructed.

3.1 State Variables

Define every state variable and include units when appropriate.

For example:

Table 1: State variables used in the model.

Variable	Meaning	Units
$S(t)$	Insert meaning here	people
$I(t)$	Insert meaning here	people
$R(t)$	Insert meaning here	people

Delete or modify this example table to match your project.

3.2 Parameters

Define every parameter.

Table 2: Model parameters.

Parameter	Meaning	Units	Value
β	Insert meaning here	Insert units	Insert value
γ	Insert meaning here	Insert units	Insert value

Explain how parameter values were selected. Possible sources include published literature, data, a reasonable hypothetical value used for exploration, or a value estimated from data.

3.3 Model Assumptions

State the assumptions used to construct the model. For example:

1. Insert assumption 1.
2. Insert assumption 2.
3. Insert assumption 3.

Explain why each assumption is reasonable for the situation being modeled.

3.4 Differential Equations

Present the model clearly. For example, a system may have the form

$$\begin{cases} x' = f(x, y), \\ y' = g(x, y). \end{cases}$$

Replace this example with your actual model.

After presenting the equations, explain **every important term**. For example, if a term

$$\beta SI$$

appears in the model, explain why it is present and what interaction it represents.

A reader should be able to understand how the assumptions lead to the equations.

3.5 Initial Conditions

State and explain the initial conditions:

$$x(0) = x_0, \quad y(0) = y_0.$$

Explain why these values were selected.

4 Mathematical Analysis

Use mathematical methods from the course to investigate the model. The exact analysis will depend on your project. You are not required to perform every item below. Choose the methods that are appropriate for your equations.

Possible analysis includes:

- finding an explicit analytical solution;
- identifying equilibrium solutions;
- determining where solutions increase or decrease;
- constructing a phase line;
- discussing stability of equilibria;
- identifying conservation laws;
- studying long-term behavior;
- deriving a threshold or critical parameter value;
- comparing two related models.

Show the important mathematical steps. Do not present a final formula without explaining how it was obtained.

4.1 Example Equilibrium Calculation

If your model has equilibria, you may organize the calculation as follows.

At equilibrium,

$$x' = 0, \quad y' = 0.$$

Therefore solve

$$f(x, y) = 0, \quad g(x, y) = 0.$$

Then interpret the resulting equilibrium in the context of your application.

Delete this example subsection if it is not appropriate for your project.

5 Numerical Simulations and Results

Use numerical simulation to investigate the model. Python and Jupyter notebooks are recommended.

Your report should normally contain at least two meaningful simulation figures or computational comparisons.

Each figure must include:

- labeled axes;
- appropriate units when available;
- a descriptive caption;
- discussion in the text explaining what the figure demonstrates.

5.1 Baseline Simulation

Describe the parameter values and initial conditions used for your baseline simulation.

Every meaningful figure must include concise alternative text in addition to its caption. The alternative text should communicate the main information conveyed by the graphic to a reader who cannot see it. In the LaTeX source, use the `alt` key of `\includegraphics`. Purely decorative graphics should instead be marked as artifacts.

Then discuss the mathematical and application-specific meaning of the figure.

5.2 Parameter or Scenario Comparison

Change one meaningful parameter, initial condition, or modeling assumption. Explain what you changed and why. Compare the new simulation with the baseline case.

Do not only state that one curve is “higher” or “lower.” Explain why the change occurs using the structure of the differential equation.

6 Discussion and Model Limitations

Interpret the mathematical and numerical results in the context of your research question. Discuss questions such as:

- What did the model predict?
- Which parameters had the strongest effect?
- Did the simulations agree with your analytical reasoning?
- Did you observe any unexpected behavior?
- What assumptions limit the model?
- What important real-world mechanisms are missing?
- How could the model be improved?

A model does not need to be perfect to be useful. Recognizing limitations is an important part of mathematical modeling and research.

7 Conclusion

Briefly summarize:

1. the research question;
2. the model used;
3. the most important mathematical result;
4. the most important simulation result;
5. the main conclusion.

Avoid introducing major new calculations in the conclusion.

Use a consistent citation style. The entries below are examples only; replace them with sources actually used in your project.

References

- [1] A. Author and B. Author, “Title of article,” *Journal Name*, vol. 1, no. 1, pp. 1–10, 2025.
- [2] C. Author, *Title of Book*. Publisher, 2024.
- [3] Organization Name, “Title of webpage or dataset,” <https://example.com>. Accessed: Month Day, Year.

A Optional Computational Appendix

You may include important code, additional figures, or supplementary calculations here. Do not place large amounts of code in the main body unless the code itself is essential to the mathematical discussion.